High Performance Computing

Report CUDA

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1 Architectures

For this project I have used two workstation, my personal one and the workstation provided by the University.

HP Envy 17-ae103nl (Laptop)

- **CPU**: Intel Core i7-6500U @ 2.50 GHz
- **Architecture**: x86_64
- **Core fisici**: 2
- **Thread logici**: 4 (Hyper-Threading)

University Workstation (Room 210)

- **CPU**: Intel Core i9-12900K (12th Gen)
- **Architecture**: x86_64
- **Core(s) per socket**: 16
- **CPU(s)**: 24
- **Thread(s) per core**: 2
- **Socket(s)**: 1
- **CPU max MHz**: 4.02 GHz
- **Profiling Tools**: Intel VTune Profiler, Intel Advisor

To use the Intel oneAPI tools, the environment was loaded with:

```
source /opt/intel/oneapi/setvars.sh
```

2 Hotspot Analysis

3 Vectorization

4 Parallelization

5 Conclusions